High Gain Low Power Low Noise Trans-impedance Amplifier for Current Sensing Application

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Abstract—**This paper work is about a transimpedance amplifier which have very high gain and sensitivity designed using cadence CMOS 0.18 µm technology with supply voltage of 1.8V dissipates power 43µW suitable for current sensing signals from a sensor, molecular and nanodevice system. The integrator based circuit followed by an ADC. Because of the ability to handle large standing currents, the circuit is competent for use in the biological system where physical medium is also present. The trans-impedance amplifier (TIA) designed has low noise high gain which make it suitable to interface with ADC. A capacitive – feedback current amplifier drives current in to active load to obtain a 80 M-Ω trans-impedance gain , 1.79 MHZ bandwidth phase response near 0.**

Keywords- TIA-Transimpedance amplifier, ADC-Analog-to-digital convertor, Op-amp- operational amplifier, MEMS-Microelectromechanical Systems, sensor current, feedback capacitor.

I. INTRODUCTION

There are many application of Current sensors in the field of image sensing , biomedical and automobile sector [1],[2]. So, the design of low power sensor interface circuits have augmented rise in last few years. The objective of this paper is to design a low power circuit that can sense current from other physical sources. A current sensor circuit have sensor element which gives an output current in proportionate to the sense quantity. Which is further given to A/D convertor. TIA based conversion technic is one of the others that convert sensor information in digital form. Because output of sensor is an analog signal either in the form of current or voltage further for digital conversion ADC is required. Output of ADC is typically next processed by a digital signal processor before acting on the information gathered by the sensor. Many techniques were propounded to efficiently transform sensor information to digital output. One of the architecture is TIA based current to digital conversion technic. TIA converts input current to output voltage which is digitized using an ADC. The feedback capacitor, C_D at high frequencies is required to filter noise. The fixed reference voltage $\mathbf{V_{bias}}$ is achieved with the help of op-amp which sensor node voltage to get desired reference voltage. The implementation of TIA-dependent interface circuit is easy to design, a high value of feedback resistor is used to sense low input current values. An amplifier with high *stability* and accuracy, low noise, is required with high resolution voltage-domain ADC [3]. Fig. 1(b) shows the architecture of the trans-impedance amplifier (TIA) [4],[5].

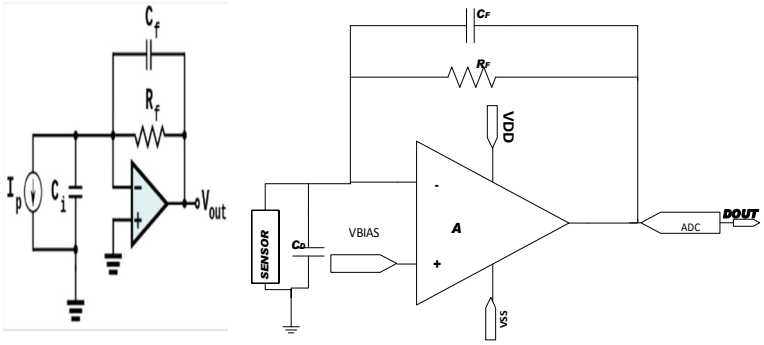


Fig. 1. Trans-impedance amplifier topology: (a) Resistive feedback TIA. (b) Trans-amplifier based ADC

II. ANALYSIS

Op-amp based TIA is used to convert current to voltage. The main purpose of using TIA is to amplify output current of Geiger–Müller tubes to useful voltage. Because of more linear response of current than voltage. It is prefer to use current to voltage converter with sensor, because current has more linear response than voltage. The current sensor has nearly about to 1% nonlinearity only. The TIA provides low impedance to the sensor and segregate it from the output voltage of the op-amp. The basic form of TIA has large value of feedback resistor, R_F. This resistor is responsible for high gain of amplifier because of its inverting configuration. For different application, different configurations of transimpedance amplifier is used. But one thing is common to all is that all have to change low sensed

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current value to a useful voltage. The TIA based op-amp has low impedance then sensor and segregates. The simplest form of TIA uses a very high valued feedback resistor, R_f. This resistor sets the amplifier gain and value of $-R_f$ is due to the inverting configuration of amplifier. Current and offsets voltage as well as gain and bandwidth change with types of sensors and different structure of transimpedance amplifiers has been used. Sensor is connected between and the inverting terminal of the op-amp, as shown in the fig 1(b). The non-inverting terminal of op-amp has been biased. This provides a low impedance load for the sensor element, which keeps the sensor element voltage low. The large gain of the op-amp helps to maintain feedback current through $\mathbf{R_f}$ and sensor current same. Because of sensor element input offset is very low in this self-biased mode. This allow a very high gain without any large output offset voltage. This configuration is used with equipment that generate very low current level and required a high gain

$$\mathbf{-I_P = \frac{V_{OUT}}{R_f}} \quad \textbf{and}$$

$$\mathbf{-R_f = \frac{V_{OUT}}{I_P}}$$

This is DC and Transimpedance amplifier having low frequency of gain. Due to high gain, output DC offset will rise if any input offset voltage at the non-inverting terminal of the op-amp will found. Similarly an output offset will be generated if inverting terminal input find bias current. Therefore TIA are generally designed with FET input op-amps which have little input offset voltages to reduce these effects[3]. Fig.1(b) C_f the feedback capacitor, is used to rise the stability of system.

Here we are working on TIA topology, which was substantiated by Razavi and shown in Fig. 1 [3].The gain set by the feedback resistor of a transimpedance amplifier and frequency response are inversely proportional to the product of the gain, Vin/Vo , for any given op-amp it is very close to be a constant. The sensors used with TIA have usually more capacitance than an op-amp can handle. Fig. 1(b) models the sensor as a current source and a capacitor, Ci. This input terminals of the op-amp has capacitance, which contain the internal capacitance of the op-amp,that have low pass filter property. Bode plot of uncompensated transimpedance amplifier the feedback path. The feedback factor $\boldsymbol{\beta}$ classify low pass response of this filter, which reduces the feedback signal. This factor play a vital role on the amplifier gain.

$$\boldsymbol{\beta} = \frac{X_{ci}}{R_f + X_{ci}} = \frac{1}{1 + R_f C_{is}}$$

Where the capacitance Ci has the reactance of X_{ci}.

The circuit's response equation is following

$$V_{out} = -\frac{I_P R_f}{1 + \frac{1}{A_{OL}\beta}}$$

While considering the response of low-pass filter effect.
Where

A_{OL} is the open loop gain of the op-amp

$\boldsymbol{\beta}$ is the feedback factor, at low frequency it has less effect on the amplifier response.

The response of amplifier would be very nearer to the ideal.

The feedback resistor is the dominant source of noise in transimpedance amplifier in most of practical cases. The voltage noise over the feedback resistance is directly proportional to output referred voltage noise. The noise has an RMS amplitude of $V_{n,or} = \sqrt{4K_B T R_f \Delta f}$. Although $\sqrt{R_f}$, increases the output noise voltage also increases proportionally yet the transimpedance has linear dependency on input referred noise current [6]:

$$\mathbf{i_{n,ir}} = \sqrt{\frac{4K_B T R_f \Delta f}{R_f}}.$$

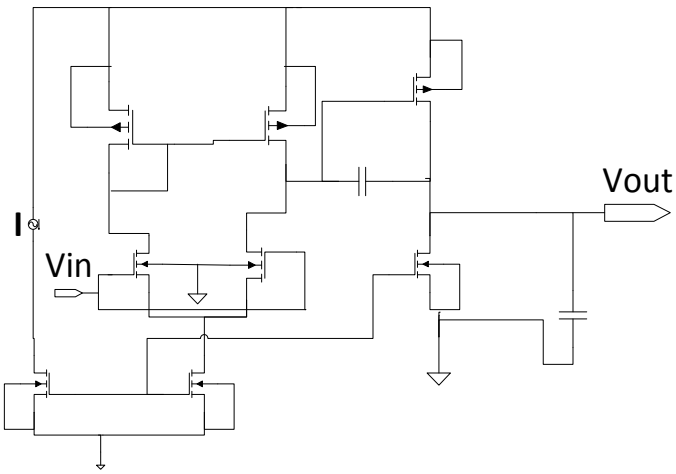


Fig. 2 schematic of high gain op-amp.

III. Simulation Result and Discussion

TIA based architecture for current sensing application required high gain to sense the low input current. So the op-amp here has 80db and 1.79 MHz gain & bandwidth respectively holds off a phase response between -3.5 and-10 from 6 kHz to 200 kHz, that is capable for

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several MEMS devices with the input current noise 26 f A/$\sqrt{Hz}$.

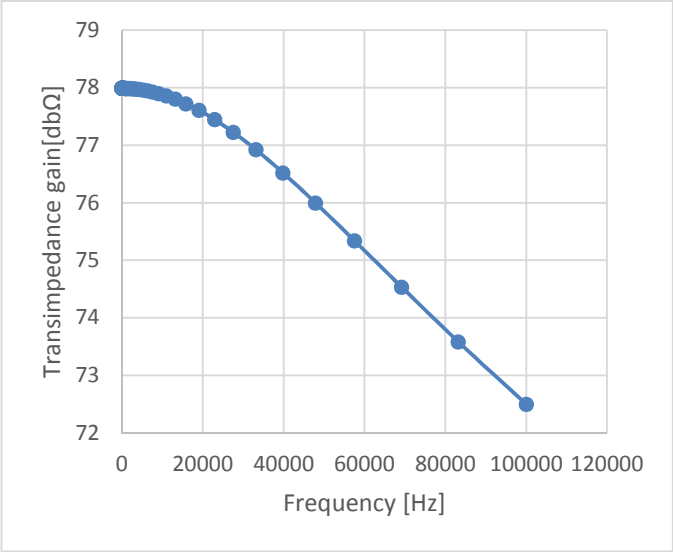


Fig.3 Measurements of the Trans-impedance gain of TIA.

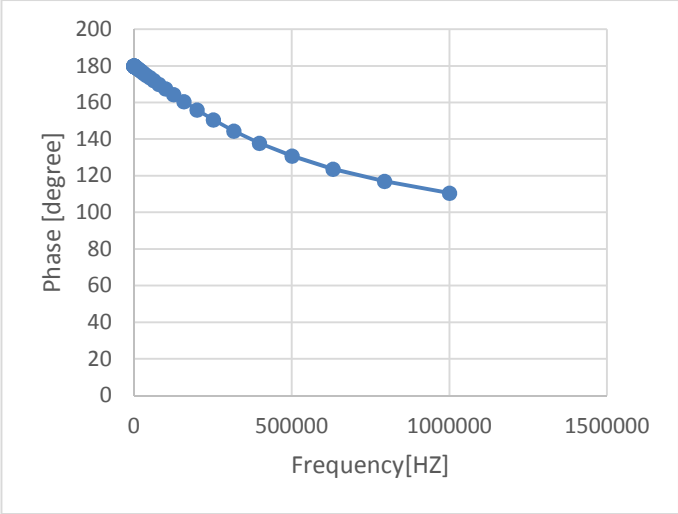


Fig.4 Measurements of the Trans-impedance phase of TIA.

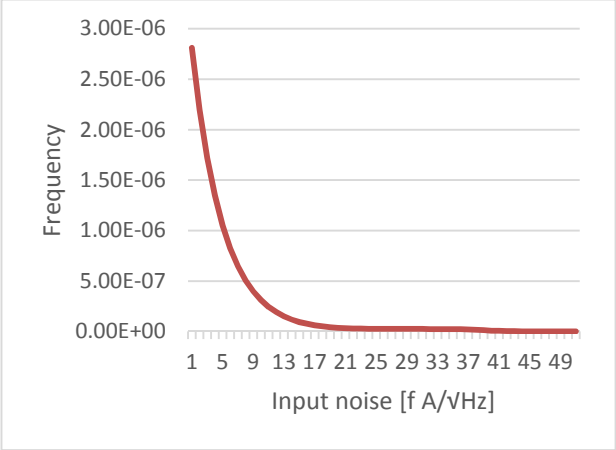


Fig.5 Input-current noise spectrum

Table 1. Shows the comparison of previously proposed architecture. The first column describe the performance of op-amp upon which TIA in this work was modelled. The reduced bandwidth of TIA here is designed but have high gain, low noise and low power.

IV. Conclusion

This paper has presented trans-impedance amplifier based current readout application which can be further digitalized using ADC architecture. This TIA is suitable for low current in the range of n A to u A. The TIA has 80db gain with bandwidth of 1.79 MHz. A phase response near zero degree between 5 kHz and 210 kHz, and 25 f A/√Hz input current noise.

TABLE 1. Comparison to Prior Architecture

	[3]	[7] [8]	[9]	This work
Gain	8.7kΩ	65MΩ	1.6MΩ	80MΩ
Bandwidth	550MHz	5 MHz	230 kHz	1.8 MHz
Power	30mW	25 mW	400 µW	46 µW
Input Noise	4.5pA/√Hz	3fA/√Hz	88fA/√Hz	25fA/√Hz
technology	0.6-µm CMOS	0.35-µm CMOS	0.6-µm CMOS	0.18-µm CMOS

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